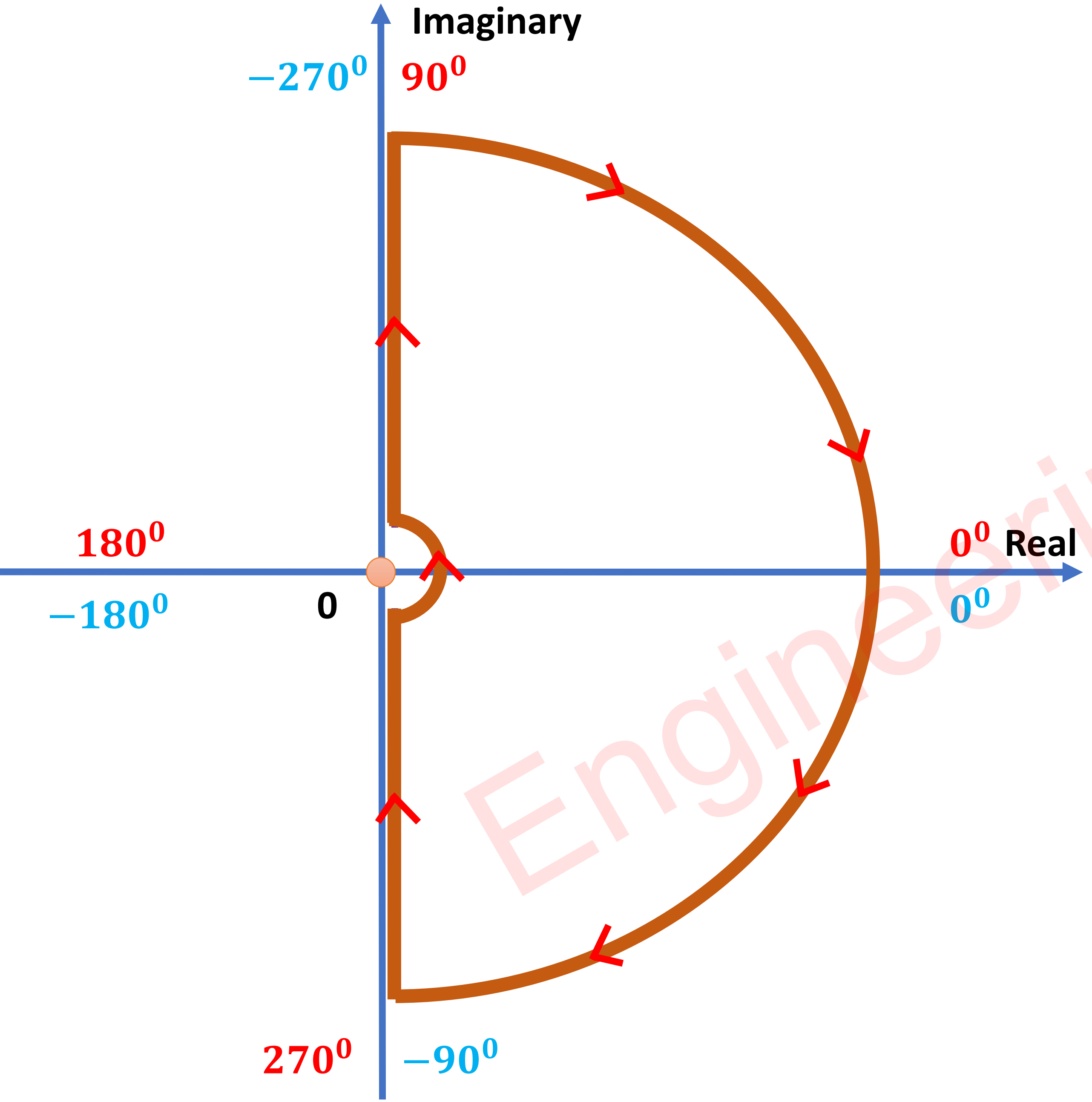
Nyquist Plot

Steps of Nyquist Plot

- Step-1: Polar Plot
- Step-2: Inverse Polar Plot {Mirror Image of Polar Plot with respect to Real Axis}
- Step-3: Nyquist Contour {Arc from $\omega = 0_-$ to $\omega = 0_+$ in clockwise direction}

Calculation of Nyquist Plot



- **Step-1: Polar Plot {Put $S = j\omega$},**
  - Plot $G(j\omega)$
- **Step-2: Inverse Polar Plot {Put $S = -j\omega$},**
  - Plot $G(-j\omega)$
- **Step-3: Put $S = \lim_{R\to\infty} Re^{j\theta}$**
  - Plot $G(Re^{j\theta})$
  - Here, $\theta \to (90^0, -90^0)$ and $R \to \infty$
- **Step-4: Put $S = \lim_{r\to 0} re^{j\theta}$**
  - Plot $G(re^{j\theta})$
  - Here, $\theta \to (-90^0, 90^0)$ and $r \to 0$
  - We need this only if we have pole at Origin.

Stability using Nyquist Plot

- ❑ Using the Polar Plot, we can identify the stability of the OLTF, but using the Nyquist Plot, we can identify the stability of OLTF and CLTF.
- ❑ Stability using Nyquist Plot is based on following equation:

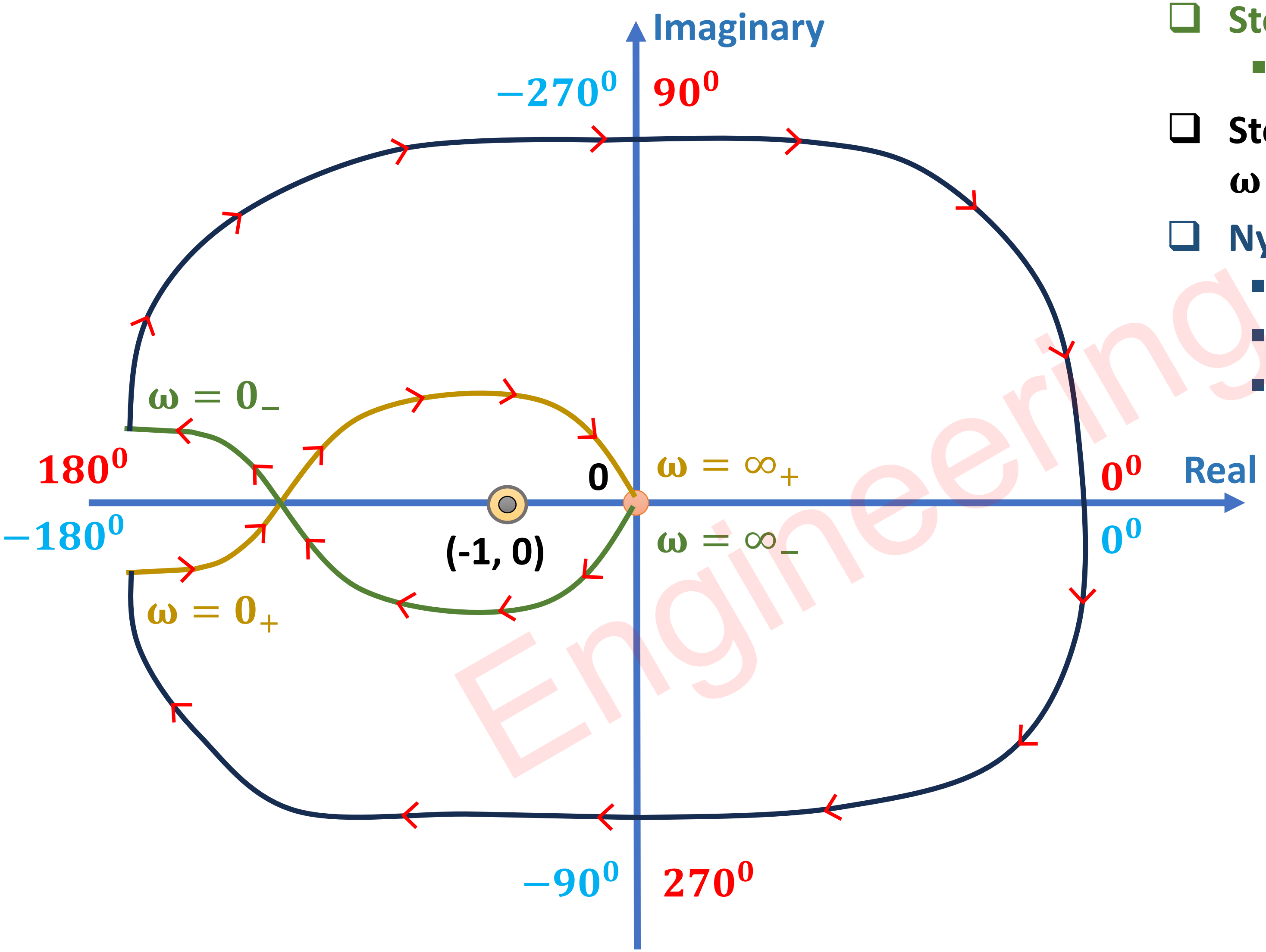
$$\Rightarrow \mathbf{N = P - Z}$$

  - Where, N = Number of encirclements around (-1, 0), positive for anticlockwise encirclements and negative for clockwise encirclements.
  - P = Open Loop Poles on RHP
  - Z = Closed Loop Poles on RHP

P	Z	Stability
$\mathbf{P = 0}$	$\mathbf{Z = 0}$	❑ Open Loop and Closed Loop Systems are stable.
$\mathbf{P \neq 0}$	$\mathbf{Z = 0}$	❑ Open Loop is unstable and Closed Loop is stable.
$\mathbf{P = 0}$	$\mathbf{Z \neq 0}$	❑ Open Loop is stable and Closed Loop is stable is unstable.
$\mathbf{P \neq 0}$	$\mathbf{Z \neq 0}$	❑ Open Loop and Closed Loop Systems are unstable.

Example 1: Identify stability for given Open Loop Transfer Function

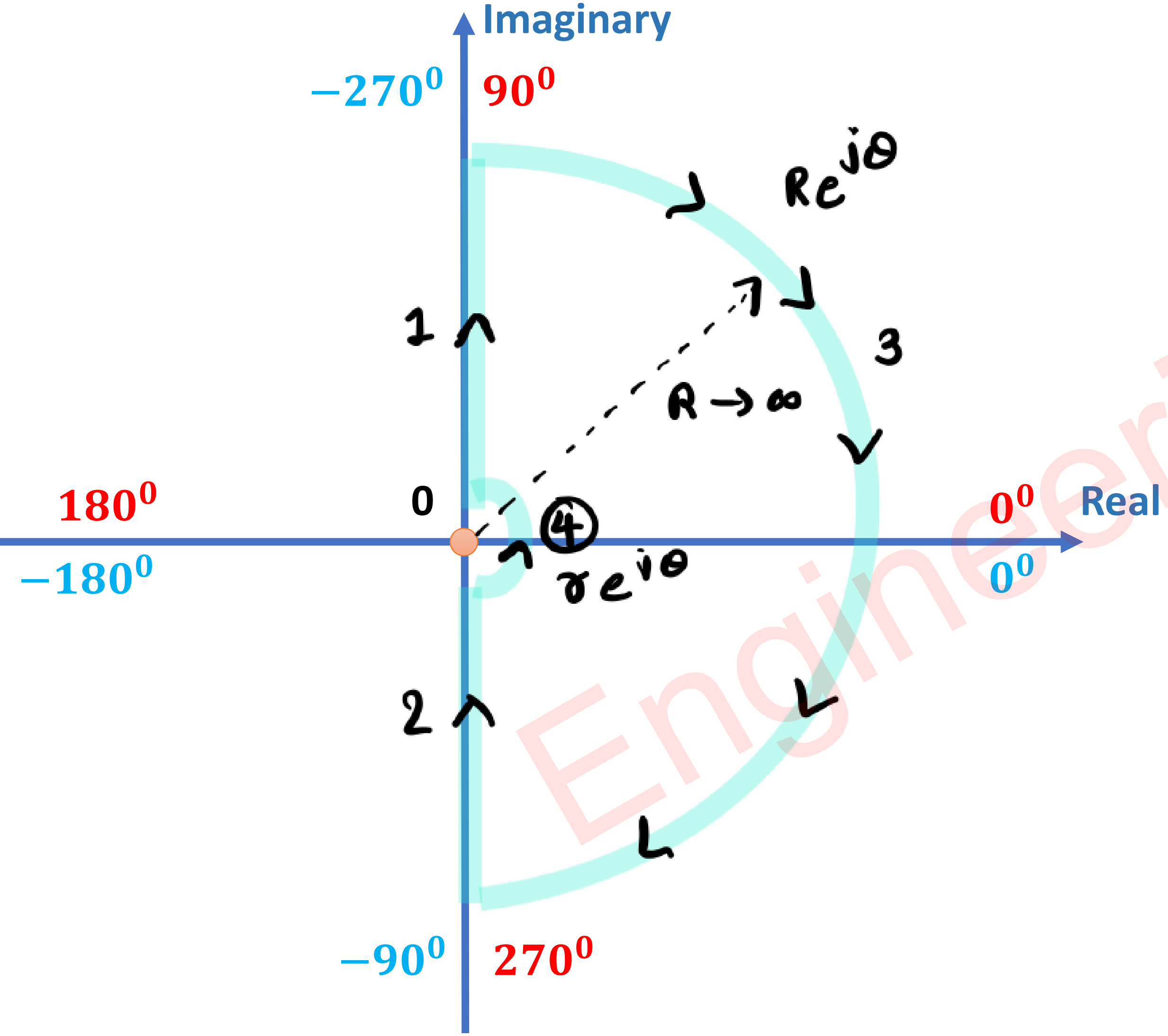
$$\Rightarrow G(S) = \frac{(4S+1)}{S^2(S+1)(2S+1)}$$



- Step-1: Polar Plot {Put $S = j\omega$},
  - Plot $G(j\omega)$
- Step-2: Inverse Polar Plot {Put $S = -j\omega$},
  - Plot $G(-j\omega)$
- Step-3: Nyquist Contour {Arc from $\omega = 0_-$ to $\omega = 0_+$ in clockwise direction}
- Nyquist Stability Equation: N = P – Z
  - N = -2 {Clockwise sign is –Ve}
  - P = 0 {Open Loop Poles on RHP} Stable
  - Hence, Z = 2 {Closed Loop Poles on RHP} Unstable

Example 2: Identify stability for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{1}{S^2(S+1)(2S+1)}$$



~ $s = j\omega \rightarrow$ Polar plot

~ $s = -j\omega \rightarrow$ Inverse Polar plot

~ $s = \lim_{R\to\infty} Re^{j\theta}$, $\theta \rightarrow 90^\circ$ to -90°

$$\Rightarrow G(s) = \lim_{R\to\infty} \frac{1}{R^2 e^{2j\theta}(1+Re^{j\theta})(1+2Re^{j\theta})}$$

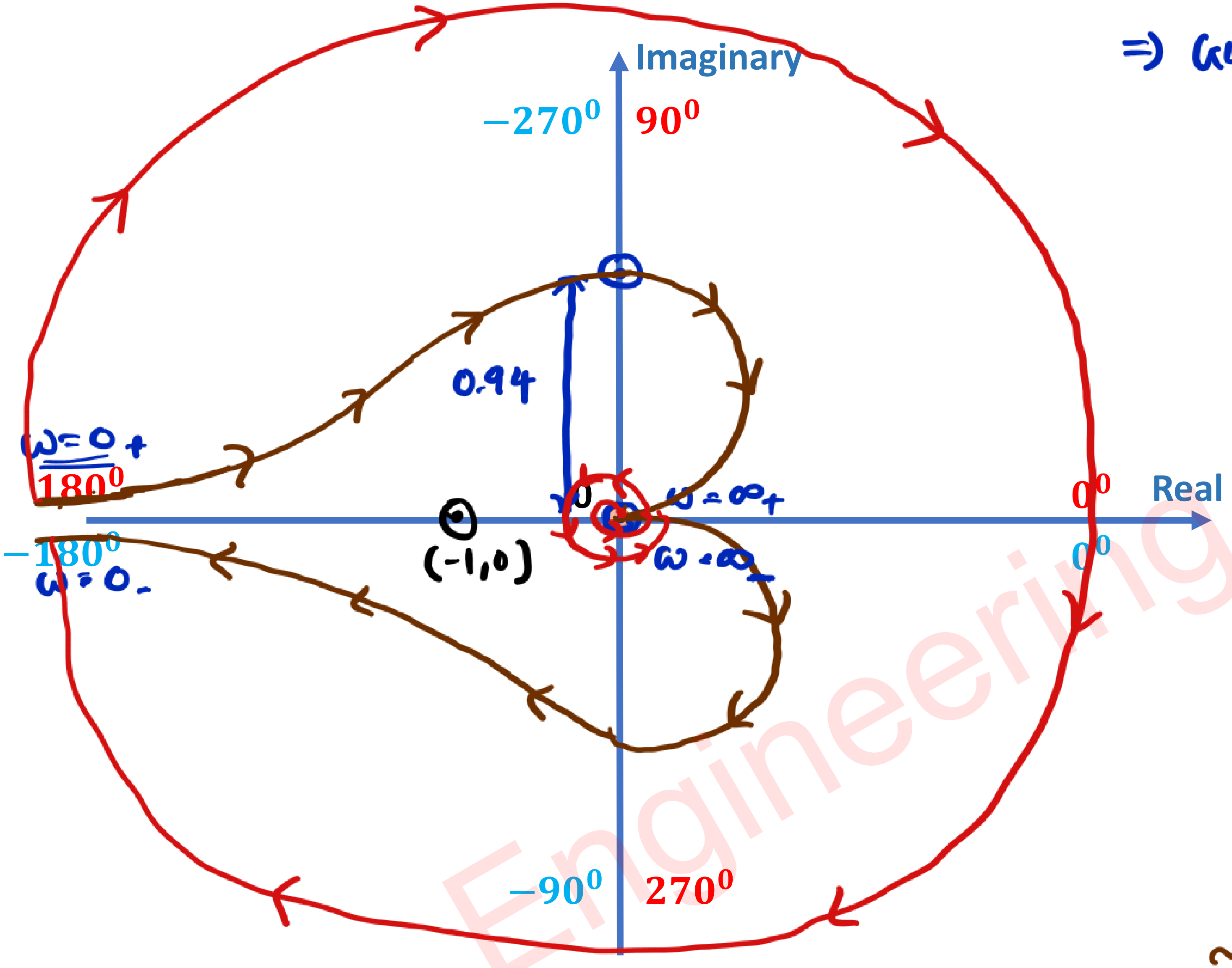
$$= \lim_{R\to\infty} \frac{1}{2R^4} e^{-4j\theta} \frac{1}{(1+1/Re^{j\theta})(1+1/2Re^{j\theta})}$$

$$= \underbrace{\frac{1}{2R^4}}_{\text{Mag}=0} \underbrace{e^{-4j\theta}}$$

Phase $\rightarrow -360^\circ$ to 360°

[two Circles in ACW]

~ $s = \lim_{r\to 0} re^{j\theta}$, $\theta \rightarrow -90^\circ$ to 90°



$$\Rightarrow G(s) = \lim_{r \to 0} \frac{1}{r^2 e^{2i\theta}(1 + re^{i\theta})(1 + 2re^{i\theta})}$$

$$= \underbrace{\frac{1}{r^2}}_{\text{Mag} = \infty} \underbrace{e^{-2i\theta}}_{\text{Phase} \to 180 \text{ to } -180^\circ}$$

[CW circle, One circle]

~ stability eqⁿ

$\Rightarrow N = P - Z$

$\Rightarrow -2 = 0 - Z$

$\boxed{\Rightarrow Z = 2}$

~ Two poles in RHP for CLTF, CLTF is unstable.

~ Polar plot, $s = j\omega$

$$\Rightarrow G(j\omega) = \frac{1}{(j\omega)^2(1+j\omega)(1+2j\omega)}$$

~ Polar Form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

- $|G(j\omega)| = \dfrac{1}{\omega^2\sqrt{1+\omega^2}\sqrt{1+4\omega^2}}$

- $\angle G(j\omega) = -180^\circ - \tan^{-1}(\omega) - \tan^{-1}(2\omega)$

~ Find magnitude & phase for $\omega=0$ & $\omega=\infty$

- At $\omega=0$, mag. $=\infty$, phase $=-180^\circ$
- At $\omega=\infty$, mag. $=0$, phase $=-360^\circ$

~ Separate Real & Imag. Component of $G(j\omega)$

$$\Rightarrow G(j\omega) = \frac{1}{(j\omega)^2(1+j\omega)(1+2j\omega)} \times \frac{(1-j\omega)(1-2j\omega)}{(1-j\omega)(1-2j\omega)}$$

$$\Rightarrow G(j\omega) = \frac{-(1-2\omega^2-3j\omega)}{\omega^2(1+\omega^2)(1+4\omega^2)}$$

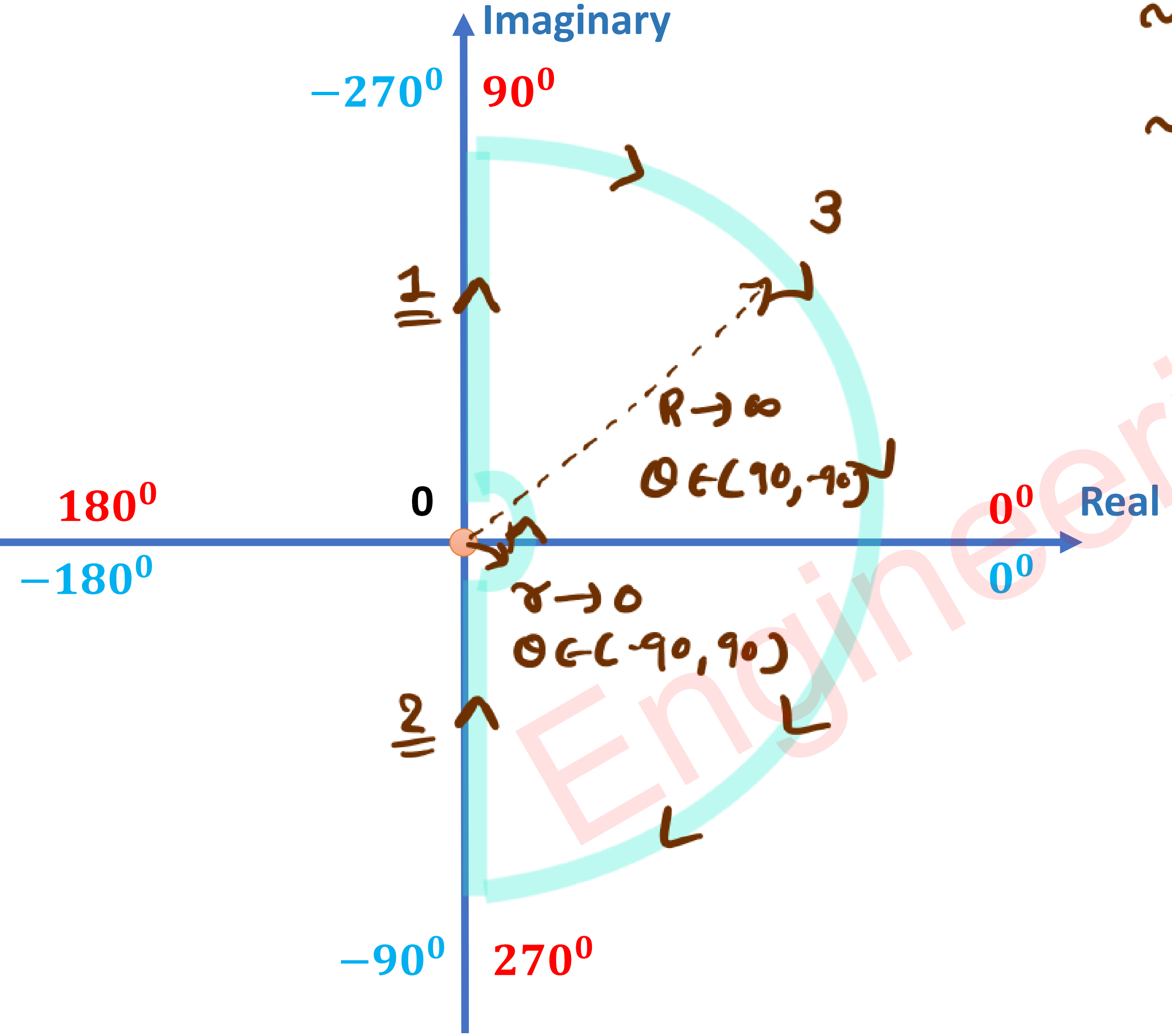
$$\Rightarrow G(j\omega) = \underbrace{\frac{2\omega^2-1}{\omega^2(1+\omega^2)(1+4\omega^2)}}_{\text{Real }[G(j\omega)]} + j\underbrace{\frac{3}{\omega(1+\omega^2)(1+4\omega^2)}}_{\text{Imag. }[G(j\omega)]}$$

- Intersection to Imag. Axis, $\Rightarrow$ Real $[G(j\omega)] = 0$

$\Rightarrow \omega = 1/\sqrt{2}$, Mag. $= 0.94$, phase $= -270^\circ$

Example 3: Identify stability for given Open Loop Transfer Function

$$\Rightarrow G(S) = \frac{1+4S}{S^2(S+1)(2S+1)}$$



~ line-1 - Polar Plot ⇒ $s = j\omega$

~ line-2 - Inverse Polar Plot ⇒ $s = -j\omega$

~ line-3 - $s = \lim_{R\to\infty} Re^{j\theta}$, $\theta \to 90$ to $-90°$

$$\Rightarrow G(s) = \lim_{R\to\infty} \frac{1+4Re^{j\theta}}{(Re^{j\theta})^2(1+Re^{j\theta})(1+2Re^{j\theta})}$$

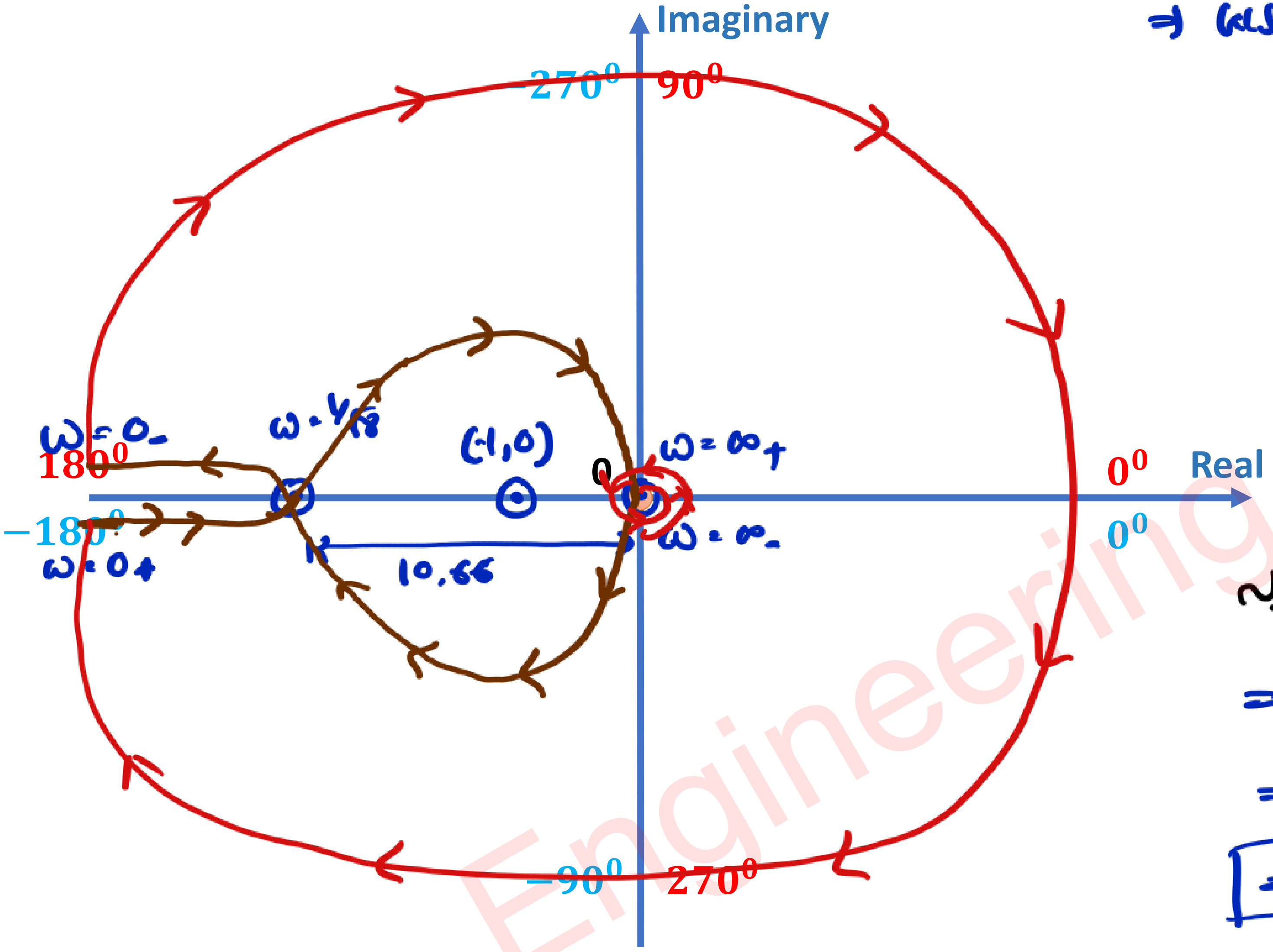
$$= \lim_{R\to\infty} \frac{2}{R^3} e^{-3j\theta} \left(\frac{1+1/4Re^{j\theta}}{(1+1/Re^{j\theta})(1+1/2Re^{j\theta})} \right)$$

$$= \underbrace{\frac{2}{R^3}}_{\text{Mag} = 0} \underbrace{e^{-3j\theta}}_{}$$

Phase → −270 to 270

[1.5 Circle, ACW]

~ line-4 - $s = \lim_{r\to 0} re^{j\theta}$, $\theta \to -90$ to 90



$$\Rightarrow G(s) = \lim_{r \to 0} \frac{1 + 4re^{i\theta}}{(re^{i\theta})^2 (1 + re^{i\theta})(1 + 2re^{i\theta})}$$

$$= \underbrace{\frac{1}{r^2}}_{\text{Mag.} = \infty} \; \underbrace{e^{-2i\theta}}_{\text{Phase} \to 180 \text{ to } -180}$$

[CW circle, one circle]

↳ Stability eqn

$\Rightarrow N = P - Z$

$\Rightarrow -2 = 0 - Z$

$\boxed{\Rightarrow Z = 2}$

↳ Two poles for CLTF in RHP,

Hence, CLTF is Unstable.

~ Polar plot , $s = j\omega$

$$\Rightarrow G(j\omega) = \frac{1 + j4\omega}{(j\omega)^2 (1 + j\omega)(1 + 2j\omega)}$$

~ Polar Form

$$\Rightarrow G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

- $|G(j\omega)| = \dfrac{\sqrt{1 + 16\omega^2}}{\omega^2 \sqrt{1 + \omega^2}\sqrt{1 + 4\omega^2}}$

- $\angle G(j\omega) = -180^\circ + \tan^{-1}(4\omega) - \tan^{-1}(\omega) - \tan^{-1}(2\omega)$

~ Find magnitude & phase for $\omega = 0$ & $\omega = \infty$

- For $\omega = 0$, mag. $= \infty$, phase $= -180^\circ$
- For $\omega = \infty$, mag. $= 0$, phase $= -270^\circ$

→ Separate Real & Imag. parts of $G(j\omega)$.

$$\Rightarrow G(j\omega) = \frac{1+j4\omega}{(j\omega)^2(1+j\omega)(1+2j\omega)} \times \frac{(1-j\omega)(1-2j\omega)}{(1-j\omega)(1-2j\omega)}$$

$$\Rightarrow G(j\omega) = \frac{-(1+j4\omega)(1-j\omega)(1-2j\omega)}{\omega^2(1+\omega^2)(1+4\omega^2)} = \frac{-(1+j4\omega)(1-2\omega^2-3j\omega)}{\omega^2(1+\omega^2)(1+4\omega^2)}$$

$$\Rightarrow G(j\omega) = \frac{-(1-2\omega^2+12\omega^2+j(4\omega(1-2\omega^2)-3\omega))}{\omega^2(1+\omega^2)(1+4\omega^2)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{-(1+10\omega^2)}{\omega^2(1+\omega^2)(1+4\omega^2)}}_{\text{Real }[G(j\omega)]} + j\underbrace{\frac{(8\omega^2-1)}{\omega(1+\omega^2)(1+4\omega^2)}}_{\text{Imag }[G(j\omega)]}$$

→ Intersection to Real Axis, ⇒ Imag. $[G(j\omega)] = 0$

$$\Rightarrow \omega = 1/\sqrt{8}, \quad \text{mag.} = 10.66, \quad \text{phase} = -180^0$$

Example 4: **Unity Feedback System has the Open Loop Transfer Function**

$$\Rightarrow G(S) = \frac{1}{(S-1)(S+2)(S+3)}$$

→ Zeros ×

→ $\underbrace{P_1 = +1}_{RHP}, \underbrace{P_2 = -2, P_3 = -3}_{LHP}$

The Nyquist Plot encircles the origin.

a. Never

b. Once ✓

c. Twice

d. Thrice

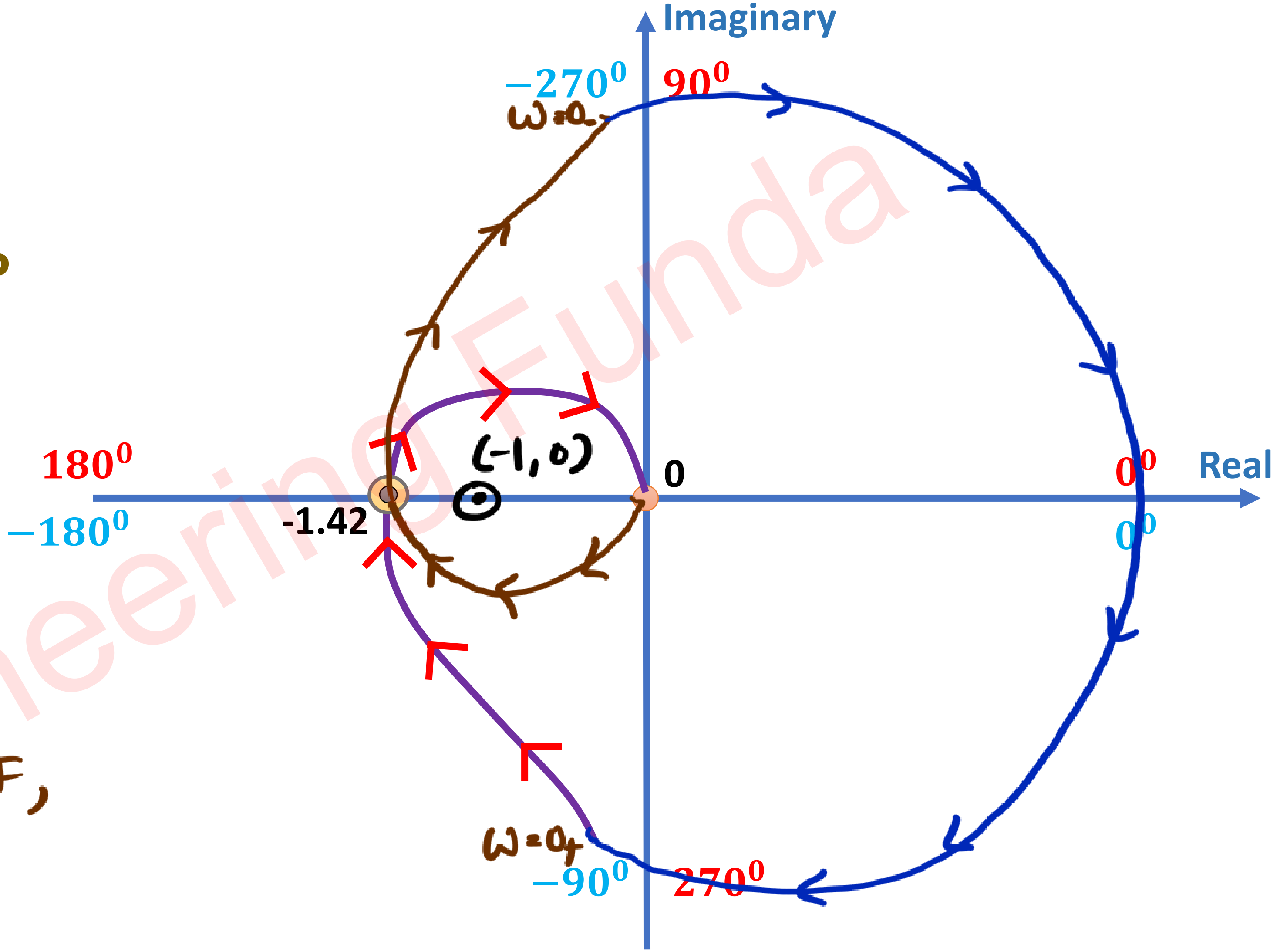
⇝ Encirclement around Origin

$\Rightarrow N = P - Z = 1 - 0 = 1$

- N = Number of encirclement around Origin (+ve ↺, -ve ↻)
- P = Number of poles in RHP for OLTF
- Z = Number of Zeros in RHP for OLTF

Example 5: The Polar Plot of an Open Loop Stable System is shown below, the closed loop system is …………

a. Always Stable
b. Marginally Stable
c. Unstable with one pole on RHP
d. ✓ Unstable with two poles on RHP



→ For stability eq.ⁿ is :

$\Rightarrow N = P - Z$

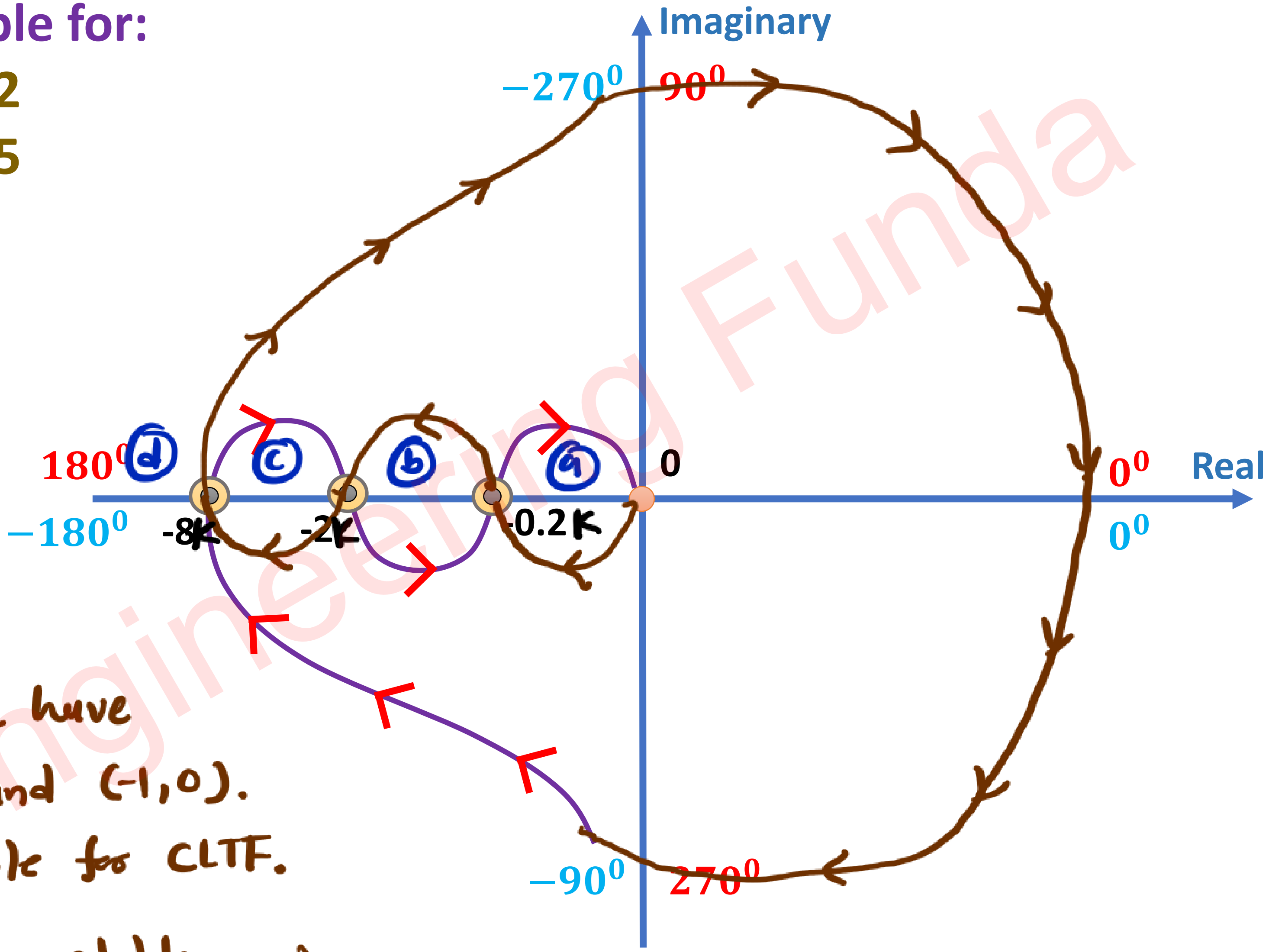
$\Rightarrow -2 = 0 - Z$

$\Rightarrow \boxed{Z = 2}$

→ Two poles in RHP for CLTF,
So system is unstable.

Example 6: The Polar diagram of a conditionally stable system for open loop gain K = 1 is shown in figure. The open loop transfer function of the system is known to be stable. The closed loop system is stable for:

a. K < 5 and 1/8 < K < 1/2
b. K < 1/8 and 1/2 < K < 5
c. K < 1/8 and 5 < K
d. K > 1/8 and 5 > K



→ Stability eqⁿ

⇒ N = P − Z

⇒ N = −Z

→ Region ⓐ & ⓒ, we have two encirclement around (−1, 0). So, system is unstable for CLTF.

→ Region ⓑ & ⓓ are stable region

- For region (b)

$\Rightarrow -2K < -1 < -0.2K$

$\Rightarrow 2K > 1$ $\Rightarrow 1 > 0.2K$

$\Rightarrow K > 1/2$ $\Rightarrow 5 > K$

$\Rightarrow \boxed{5 > K > 1/2}$

- For region (d)

$\Rightarrow -1 < -8K$

$\Rightarrow 1 > 8K$

$\Rightarrow \boxed{1/8 > K}$

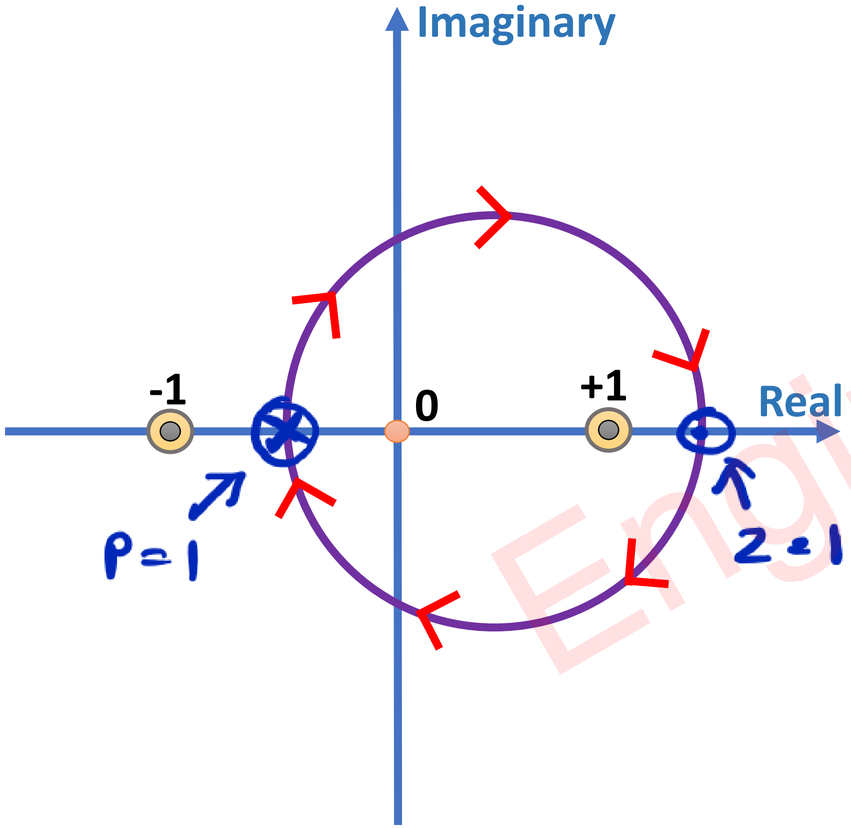
Example 7: The Nyquist Plot of G(S) is shown in figure, which one of the following conclusions is correct?

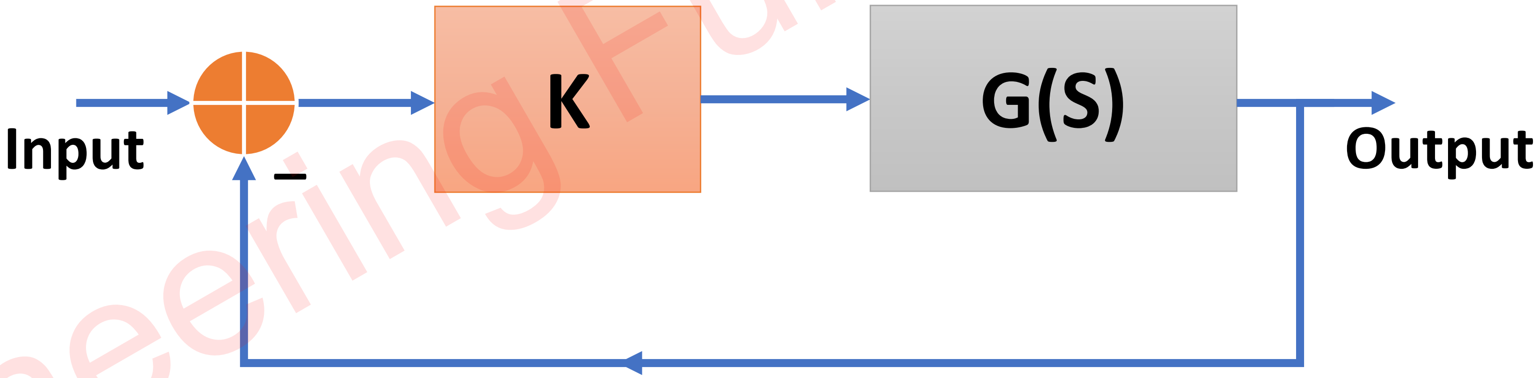
a. G(S) is an all-pass filter X

b. G(S) is strictly a proper transfer function X

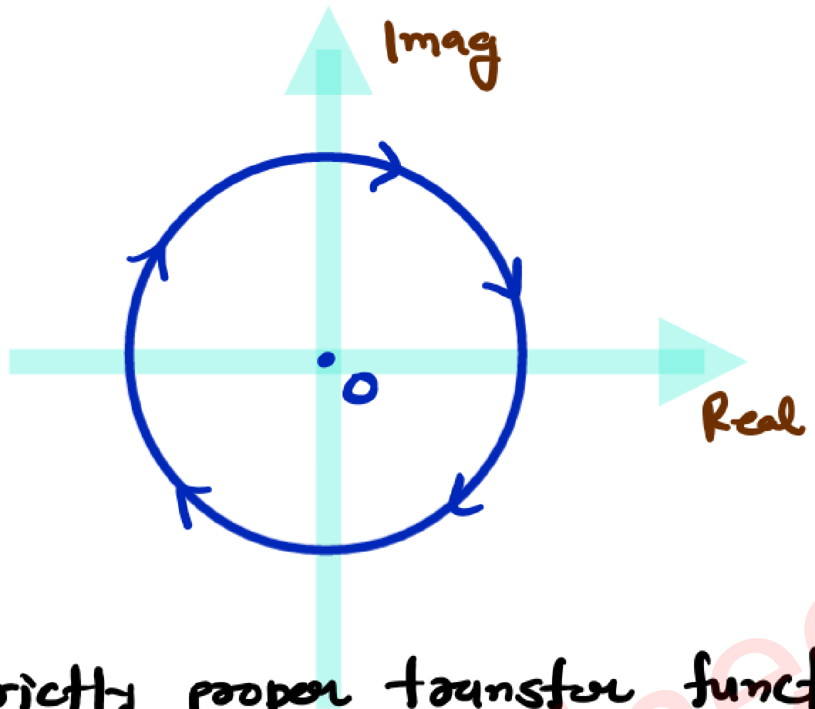
c. G(S) is a stable and MPTF X

d. The Closed loop system is unstable for large K ✓





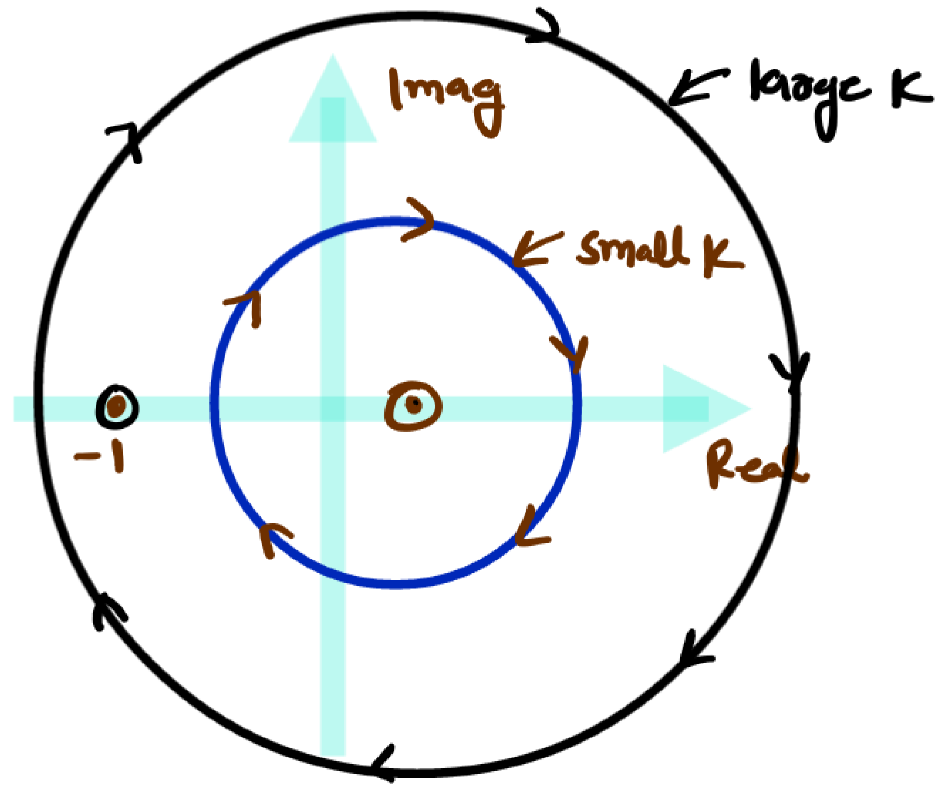
→ For all pass Filter, Magnitude should be same for all Freq.



→ In Strictly proper transfer function,

$$\underline{\underline{P > Z}}$$

→ For Minimum Phase transfer Function,
all the poles & zeros should be there
is LHP.



- $N = P - Z$

$N \rightarrow$ For large K $N = -1$; $P \rightarrow 0$

$\Rightarrow -1 = 0 - Z$

$\Rightarrow \boxed{Z = 1}$

One pole in RHP for CLTF.

Example 8: An Open Loop Transfer Function is given by:

$$\Rightarrow G(S) = \frac{K}{(S^2 + 2S + 2)(S + 2)}$$

If K = 10, the Nyquist plot does not enclose (-1, 0).

If K = 100, the Nyquist plot encloses (-1, 0).

Then Closed Loop System is ………………

a. Stable for K = 10 and K = 100.
b. Unstable for K = 10 and K = 100.
c. Stable for K = 10 and Unstable for K = 100
d. Unstable for K = 10 and Stable for K = 100

→ Stability eqⁿ

→ $N = P - Z$

→ $P_1 = -2$

$P_2 = -1 + j$

$P_3 = -1 - j$

– Ve real Value means all poles in LHP. $\boxed{P = 0}$

→ For K = 10

– $N = 0$

– $Z = 0$

– Stable CLTF

→ For K = 100

– $N \neq 0$

– $Z \neq 0$

– Unstable CLTF

Example 9: If $X = \text{Real}\,[G(j\omega)]$ and $Y = \text{Imag}\,[G(j\omega)]$, then for $\omega \to 0_+$

$$\Rightarrow G(S) = \frac{1}{S(S+1)(S+2)}$$

What is value of X and Y?

$\rightsquigarrow$ Put $s = j\omega$

$$\Rightarrow G(j\omega) = \frac{1}{j\omega(1+j\omega)(2+j\omega)} \times \frac{(1-j\omega)(2-j\omega)}{(1-j\omega)(2-j\omega)}$$

$$\Rightarrow G(j\omega) = \frac{-j(2-\omega^2-3j\omega)}{\omega(1+\omega^2)(4+\omega^2)}$$

$$\Rightarrow G(j\omega) = \underbrace{\frac{-3\omega}{\omega(1+\omega^2)(4+\omega^2)}}_{\text{Real } G(j\omega)} + j\underbrace{\frac{(\omega^2-2)}{\omega(1+\omega^2)(4+\omega^2)}}_{\text{Imag } G(j\omega)}$$

$$- X = \lim_{\omega \to 0_+} \text{Real } G(j\omega) = \frac{-3}{4} \qquad - Y = \lim_{\omega \to 0_+} \text{Imag } G(j\omega) = \infty$$

Parameters of System

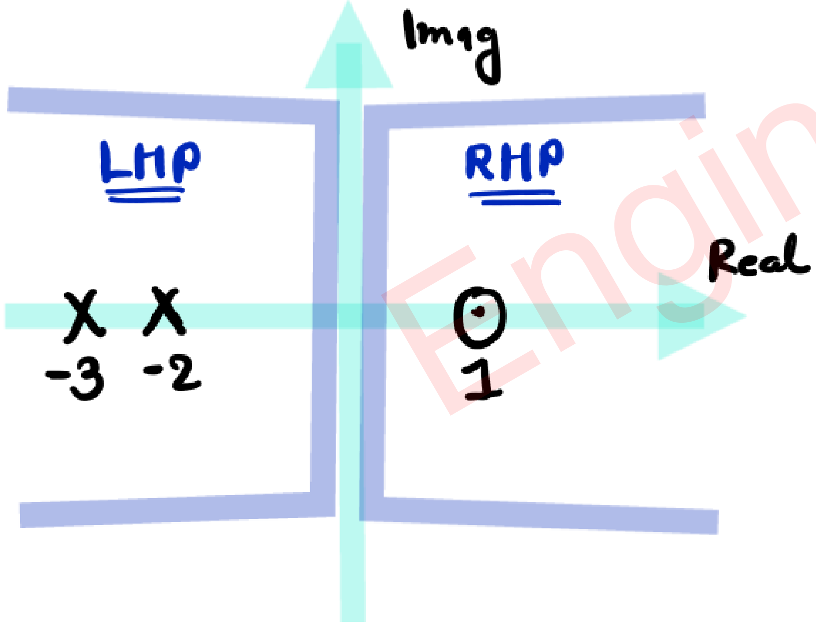
Example 1: An Open Loop System represented by the transfer function:

$$\Rightarrow G(S) = \frac{S-1}{(S+2)(S+3)}$$

→ $P_1 = -2$, $P_2 = -3$

→ $z_1 = 1$

A. Stable and of the minimum phase type
B. ✓ Stable and of the non minimum phase type
C. Unstable and of the minimum phase type
D. Unstable and of the non minimum phase type



→ All poles in LHP → Stable System

→ One Zero in RHP → Non-Minimum phase type system.

Example 2: **The Phase Margin of the system for which Loop Gain is given by:**

$$\Rightarrow GH(j\omega) = \frac{1}{(1+j\omega)^3}$$

A. $-\pi$

B. π ✓

C. 0

D. $\pi/2$

$\phi = -3\tan^{-1}(\omega)$

$m = \frac{1}{(\sqrt{1+\omega^2})^3}$

– $\omega_{gc} \rightarrow m = 1$

$\Rightarrow 1 = \frac{1}{(\sqrt{1+\omega^2})^3}$

$\Rightarrow 1+\omega^2 = 1$

$\Rightarrow \omega^2 = 0$

$\Rightarrow \omega_{gc} = 0$ rad/sec

– $\phi_{gc} = -3\tan^{-1}(\omega_{gc})$

$= 0$

– $PM = 180^\circ + \phi_{gc}$

$= 180^\circ$

$= \pi$

Example 3: The Open Loop Transfer Function of Unity Feedback Control System is given by:

$$\Rightarrow G(S) = \frac{aS+1}{S^2} \qquad \Rightarrow G(j\omega) = \frac{a(j\omega)+1}{(j\omega)^2}$$

The value of 'a' to give phase margin of 45^0 is ……………

A. 0.141
B. 0.441
C. 0.841
D. 1.141

$$\phi = -180 + \tan^{-1}(a\omega)$$

$$m = \frac{\sqrt{1+a^2\omega^2}}{\omega^2}$$

- $PM = 180^\circ + \phi_{gc} \longrightarrow 45^\circ = 180^\circ + (-180 + \tan^{-1}(a\omega_{gc}))$

$\Rightarrow 45^\circ = \tan^{-1}(a\omega_{gc})$

$\Rightarrow \boxed{1 = a\omega_{gc}}$ — 1

- For $\omega_{gc} \longrightarrow m = 1$

$$\Rightarrow 1 = \frac{\sqrt{1+a^2\omega_{gc}^2}}{\omega_{gc}^2}$$

$$\Rightarrow 1 = \frac{\sqrt{2}}{\omega_{gc}^2} \Rightarrow \boxed{\omega_{gc} = 2^{1/4}}$$ — 2

→ Put 2 into 1

$\Rightarrow 1 = a(2^{1/4})$

$\Rightarrow a = 2^{-1/4}$

$\Rightarrow \boxed{a = 0.84}$

Example 4: The Open Loop Transfer Function of a feedback control system is given by:

$$\Rightarrow G(S) = \frac{1}{(S+1)^3} \quad \Rightarrow G(j\omega) = \frac{1}{(1+j\omega)^3}$$

The Gain Margin of the system is ……………..

A. 16
B. 8
C. 4
D. 2

$$-\phi = -3\tan^{-1}(\omega)$$

$$m = \frac{1}{\left(\sqrt{1+\omega^2}\right)^3}$$

$\rightsquigarrow$ at $\omega_{pc} \rightarrow \phi = -180^\circ$

$$\Rightarrow -180^\circ = -3\tan^{-1}(\omega_{pc})$$

$$\Rightarrow 60^\circ = \tan^{-1}(\omega_{pc})$$

$$\Rightarrow \omega_{pc} = \sqrt{3} \text{ rad/sec}$$

$$\rightsquigarrow m_{pc} = \frac{1}{\left(\sqrt{1+\omega_{pc}^2}\right)^3} = \frac{1}{(\sqrt{4})^3} = \frac{1}{8}$$

$$\rightsquigarrow GM = \frac{1}{m_{pc}} = 8 = 20\log 8$$

Example 5: The Frequency response of a Linear System $G(j\omega)$ is provided in the tabular form below, The gain margin & phase margin of the system is ……………….

A. $6\text{ dB}, 30^0$ ✓
B. $6\text{ dB}, -30^0$
C. $-6\text{ dB}, 30^0$
D. $-6\text{ dB}, -30^0$

$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
1.3	-130^0
1.2	-140^0
1	-150^0 ← ω_{gc}, $\phi_{gc} = -150^\circ$
0.8	-160^0
0.5	-180^0 ← ω_{pc}, $m_{pc} = 0.5$
0.3	-200^0

$$\rightsquigarrow GM = \frac{1}{m_{pc}} = \frac{1}{0.5} = 2 = 20\log 2 = 6\text{ dB}$$

$$\rightsquigarrow PM = 180^\circ + \phi_{gc} = 180^\circ + (-150^\circ) = 30^\circ$$

Example 6: A Unity feedback system with open loop transfer function is given by:

$$\Rightarrow G(S) = \frac{1}{S(S+2)(S+4)} \Rightarrow G(j\omega) = \frac{1}{j\omega(2+j\omega)(4+j\omega)}$$

The Gain Margin of the system is ……………. dB

$\rightsquigarrow GM = \frac{1}{M_{PC}}$

$\rightsquigarrow \phi = -90 - \tan^{-1}(\omega/2) - \tan^{-1}(\omega/4)$

$M = \frac{1}{\omega\sqrt{4+\omega^2}\sqrt{16+\omega^2}}$

$\rightsquigarrow \omega_{PC} \longrightarrow \phi = -180^\circ = -90 - \tan^{-1}(\omega/2) - \tan^{-1}(\omega/4)$

$\Rightarrow +90 = \tan^{-1}\left(\frac{\omega/2 + \omega/4}{1-\omega^2/8}\right)$

$\Rightarrow \infty = \frac{\omega/2 + \omega/8}{1-\omega^2/8}$

$\Rightarrow \omega_{PC} = \sqrt{8}$ Rad/Sec

$M_{PC} = \frac{1}{\omega_{PC}\sqrt{4+\omega_{PC}^2}\sqrt{16+\omega_{PC}^2}}$

$= \frac{1}{\sqrt{8}\sqrt{12}\sqrt{24}}$

$= 48$

$\rightsquigarrow GM = \frac{1}{M_{PC}} = \frac{1}{48} = 20\log\left(\frac{1}{48}\right)$

$= -33.62$ dB

Example 7: In GH(S) plane, the Nyquist plot of the loop transfer function is given by:

$$\Rightarrow GH(S) = \frac{\pi e^{-0.25S}}{S} \qquad \Rightarrow G(j\omega) = \frac{\pi e^{-0.25(j\omega)}}{j\omega}$$

It is passes through the negative real axis at the point …………

A. (-0.25, j0)

B. ✓ (-0.5, j0)

C. (-1, j0)

D. (-2, j0)

$- \phi = -90 - 0.25\omega$

$m = \frac{\pi}{\omega}$

$- \phi = -180^\circ = -90 - 0.25\omega$

$\Rightarrow -90^\circ = -0.25\omega$

$\Rightarrow \omega = 360^\circ = 2\pi$ rad/sec

$- m = \frac{\pi}{2\pi} = \frac{1}{2}$

Example 8: In GH(S) plane, the Nyquist plot of the loop transfer function is given by:

$$\Rightarrow GH(S) = \frac{10e^{-LS}}{S} \quad \Rightarrow G(j\omega) = \frac{10\,e^{-L(j\omega)}}{j\omega}$$

The phase cross-over frequency is 5rad/sec. The value of L is ……………

A. $\pi/20$

B. $\pi/10$ ✓

C. $-\pi/20$

D. Zero

$\rightarrow \phi = -90 - L\omega$

$m = \dfrac{10}{\omega}$

$\rightarrow \omega_{pc} = 5$ rad/sec $\rightarrow \phi = -180^\circ = -90 - L(5)$

$\Rightarrow L = \dfrac{90^\circ}{5} = \dfrac{\pi}{10}$

Example 9: A system with zero initial condition has a closed transfer function given by:

$$\Longrightarrow T(S) = \frac{S^2 + 4}{(S+1)(S+4)} \Rightarrow T(j\omega) = \frac{(j\omega)^2 + 4}{(1+j\omega)(4+j\omega)} = \frac{4-\omega^2}{(1+j\omega)(4+j\omega)}$$

The system output is zero at the frequency of ....

A. 0.5 rad/sec

B. 1 rad/sec

C. 2 rad/sec ✓

D. 4 rad/sec

$$\rightarrow M = \frac{4-\omega^2}{\sqrt{1+\omega^2}\sqrt{16+\omega^2}} = 0$$

$$\Rightarrow 4-\omega^2 = 0$$

$$\Rightarrow \omega = 2 \text{ rad/sec}$$